

Injury-Aware Exercise Guidelines

General Principles

Sharp, localized, or worsening pain during a movement is a stop signal - modify or avoid the movement and consult a medical professional for persistent or severe pain.

Mild, diffuse muscle soreness (DOMS) is generally normal and different from joint or sharp pain.

This information is educational only and does not replace assessment by a physician, physiotherapist, or certified athletic trainer.

Lower Back Sensitivity

Avoid / Caution	Safer Alternatives	Notes
Loaded spinal flexion (sit-ups, heavy good mornings)	Bird-dog, dead bug, plank variations	Prioritize core bracing over flexion
Heavy conventional deadlift if pain present	Trap bar deadlift, hip thrust, glute bridge	Trap bar reduces lumbar shear load
Deep barbell back squat with flexion under load	Box squat, goblet squat, leg press	Reduce range of motion/load until pain subsides

Knee Sensitivity

Avoid / Caution	Safer Alternatives	Notes
Deep knee flexion under heavy load	Partial-range squats, wall sits, shallow step-ups	Reduce range of motion, control tempo
High-impact plyometrics	Cycling, swimming, leg press	Lower impact reduces joint stress
Running on hard surfaces with patellar pain	Elliptical, aqua jogging, cycling	Cross-training maintains cardio fitness

Shoulder Sensitivity

Avoid / Caution	Safer Alternatives	Notes
Behind-the-neck press or pulldown	Front overhead press, lat pulldown to chest	Reduces impingement risk
Heavy bench press with flared elbows	Neutral-grip dumbbell press, floor press	Floor press limits range and strain
Upright rows with heavy load	Face pulls, cable lateral raise	Face pulls strengthen rotator cuff safely

Wrist / Elbow Sensitivity

Avoid / Caution	Safer Alternatives	Notes
Straight-bar curls with heavy load	EZ-bar curls, neutral-grip hammer curls	Reduces wrist and elbow torque
Push-ups on flat wrists	Push-up handles, fists, or dumbbells	Keeps wrist in neutral position

Return-to-Training Guidance

Begin with pain-free range of motion only, then gradually reintroduce load once comfortable across a full session.

Progress load or volume by no more than about 10% per week when returning from injury.

If a modification still causes pain, remove the movement entirely and consult a professional.